NETWORKWALKS-B082

Multi-Operating-System Virtual Network and Cybersecurity Testing Lab

Author: Muhammad Kamil

Classification: Confidential

Version: 1.0

Date

June 24, 2026

Table of Contents

Task Introduction & Objectives *1*

OBJECTIVES 1

TASK SCOPE 2

TASK DELIVERABLES 2

Host System & Hardware Specifications *3*

HOST SYSTEM OVERVIEW 3

HOST HARDWARE SPECIFICATIONS 3

VIRTUALIZATION ENVIRONMENT 3

RESOURCE MANAGEMENT 4

EVIDENCE AND POCs 4

Virtualization Environment *6*

INTRODUCTION 6

VIRTUALIZATION PLATFORM 6

VIRTUAL MACHINE INVENTORY 7

VIRTUAL MACHINE MANAGEMENT 7

ISOLATION AND RESOURCE MANAGEMENT 7

EVIDENCE AND POCs 8

Network Architecture & Topology *9*

NETWORK OVERVIEW 9

IP ADDRESS ALLOCATION 9

NETWORK TOPOLOGY DIAGRAM 10

EVIDENCE AND POCs 10

Virtual Machine Configuration & Operating System Deployment *13*

OVERVIEW 13

VIRTUAL MACHINE INVENTORY 14

OPERATING SYSTEM DEPLOYMENT DETAILS 14

Conclusion *19*

Learning Outcomes *20*

References *20*

OFFICIAL DOCUMENTATION 20

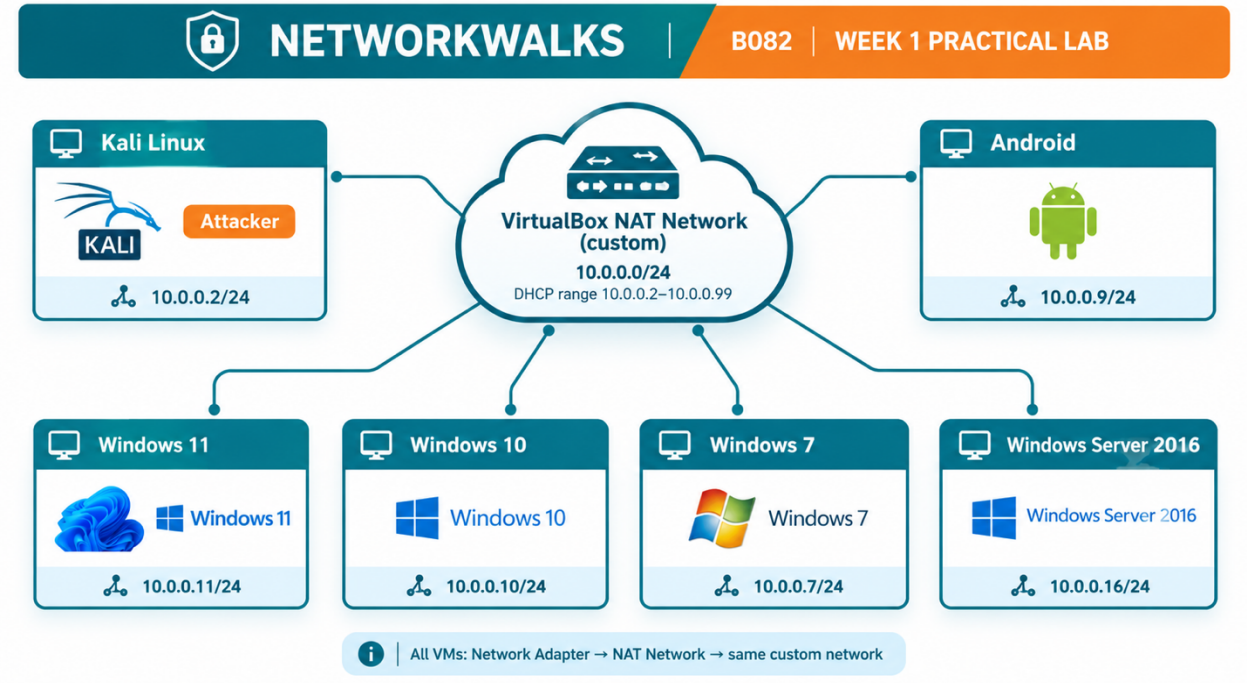
Task Introduction & Objectives

OVERVIEW

This section of document contains a virtualized multi-operating-system laboratory developed using Oracle VirtualBox. It provides an isolated network environment where multiple operating systems communicate through a shared NAT Network.

The lab is designed to provide practical experience in computer networking, operating system administration, network discovery, remote access, and cybersecurity testing.

The environment includes Kali Linux, Windows operating systems, Windows Server 2016, and Android-x86 9.0.



OBJECTIVES

The primary objectives are:

1. Build a multi-OS virtual environment using Oracle VirtualBox.
2. Configure static IP addresses for individual virtual machines.
3. Establish communication between different operating systems.
4. Practice network connectivity and reachability testing.
5. Perform network discovery and service enumeration using Nmap.

6. Configure remote Android access using Android Debug Bridge (ADB).

7. Develop practical skills in network administration and cybersecurity assessment

TASK SCOPE

The project covers the installation, configuration, and testing of six virtual machines.

Operating system	**Primary purpose**
Kali Linux	Network scanning and security testing
Windows 7	Legacy OS testing
Android-x86 9.0	Android testing and ADB
Windows 10	Windows client testing
Windows 11	Modern Windows client testing
Windows Server 2016	Server administration and testing

TASK DELIVERABLES

- Six configured virtual machines.
- A shared 10.0.0.0/24 virtual network.
- Network connectivity verification evidence.
- Nmap discovery and enumeration results.
- Android ADB remote-access configuration.
- Technical documentation and screenshots.

Host System & Hardware Specifications

HOST SYSTEM OVERVIEW

The lab is hosted on an Intel-based Apple MacBook Pro (2019), which provides the hardware resources required to operate multiple virtual machines through Oracle VirtualBox.

The host system serves as the primary virtualization platform, managing virtual hardware, storage, memory allocation, and network connectivity.

HOST HARDWARE SPECIFICATIONS

Component	Specification
Host device	Apple MacBook Pro (2019)
Processor architecture	Intel x86_64
RAM	16 GB
Available storage	Previously reported as 100+ GB free
Host operating system	macOS
Virtualization software	Oracle VirtualBox
VirtualBox version	7.2.6 r172322

VIRTUALIZATION ENVIRONMENT

Oracle VirtualBox is used to create and manage the six virtual machines.

The following resources are allocated according to each guest operating system's requirements:

- Virtual CPUs for processing.
- Virtual RAM for system operations.
- Virtual hard disks for operating system installation and storage.
- Virtual network adapters for communication.
- Virtual display and input devices.

RESOURCE MANAGEMENT

The host system has 16 GB of physical RAM. Therefore, memory allocation must be managed carefully when running multiple virtual machines simultaneously.

The laboratory uses individual VM configurations to balance performance and resource utilization.

Resource management considerations:

1. Avoid running all six VMs simultaneously unless necessary.
2. Allocate RAM according to each operating system's requirements.
3. Monitor available host storage.
4. Shut down unused virtual machines.
5. Maintain sufficient resources for macOS

EVIDENCE AND POCs

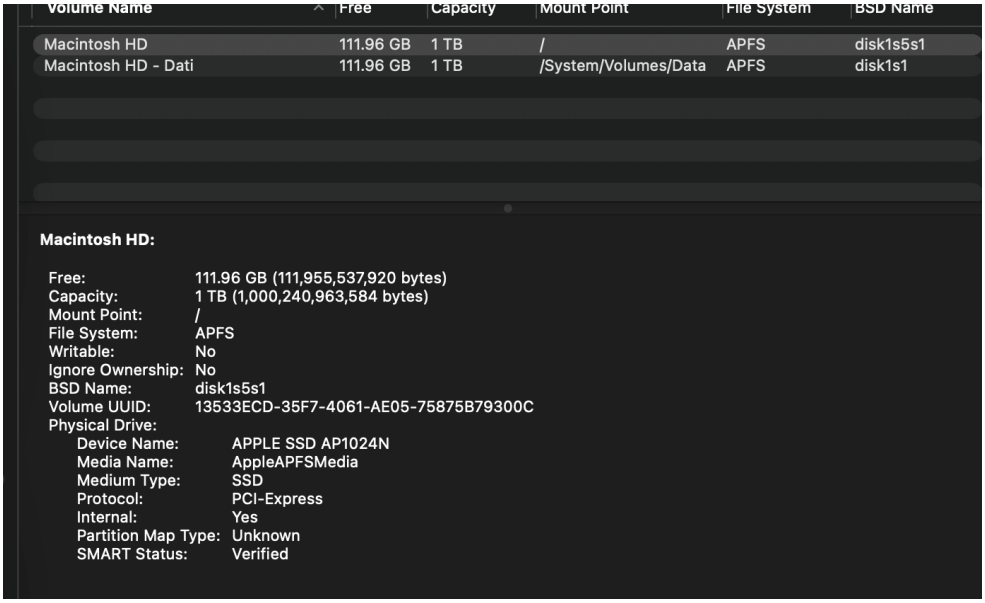
Host System Information:



VirtualBox Version:



Available Storage:



The Intel-based MacBook Pro provides the host environment for the NETWORKWALKS-B082 laboratory. With 16 GB RAM and Oracle VirtualBox 7.2.6, it supports the deployment and management of multiple operating systems for practical networking and cybersecurity exercises.

Virtualization Environment

INTRODUCTION

Virtualization is the process of creating virtual computing environments that operate independently of the physical host system.

For this task, Oracle VirtualBox 7.2.6 is used as the primary hypervisor to deploy, configure, and manage multiple guest operating systems on an Intel-based MacBook Pro.

This approach allows different operating systems to operate within a shared virtual networking environment without requiring separate physical computers.

VIRTUALIZATION PLATFORM

Component	**Configuration**
Hypervisor	Oracle VirtualBox
Version	7.2.6 r172322
Host architecture	Intel x86_64
Host RAM	16 GB
Virtual disk format	VDI
Disk allocation	Dynamically allocated
Network mode	NAT Network
Network name	NATNetwork

VIRTUAL MACHINE INVENTORY

The laboratory consists of six guest operating systems.

VM	Guest operating system	Role
VM-01	Kali Linux	Attacker / testing machine
VM-02	Windows 7	Legacy Windows environment
VM-03	Android-x86 9.0-r2	Android testing target
VM-04	Windows 10	Windows client
VM-05	Windows 11	Modern Windows client
VM-06	Windows Server 2016	Server environment

VIRTUAL MACHINE MANAGEMENT

VirtualBox provides several management features used throughout the project:

- Start and shut down virtual machines.
- Modify CPU and RAM allocations.
- Attach ISO installation media.
- Configure virtual network adapters.
- Manage virtual storage devices.
- Monitor guest operating system status.

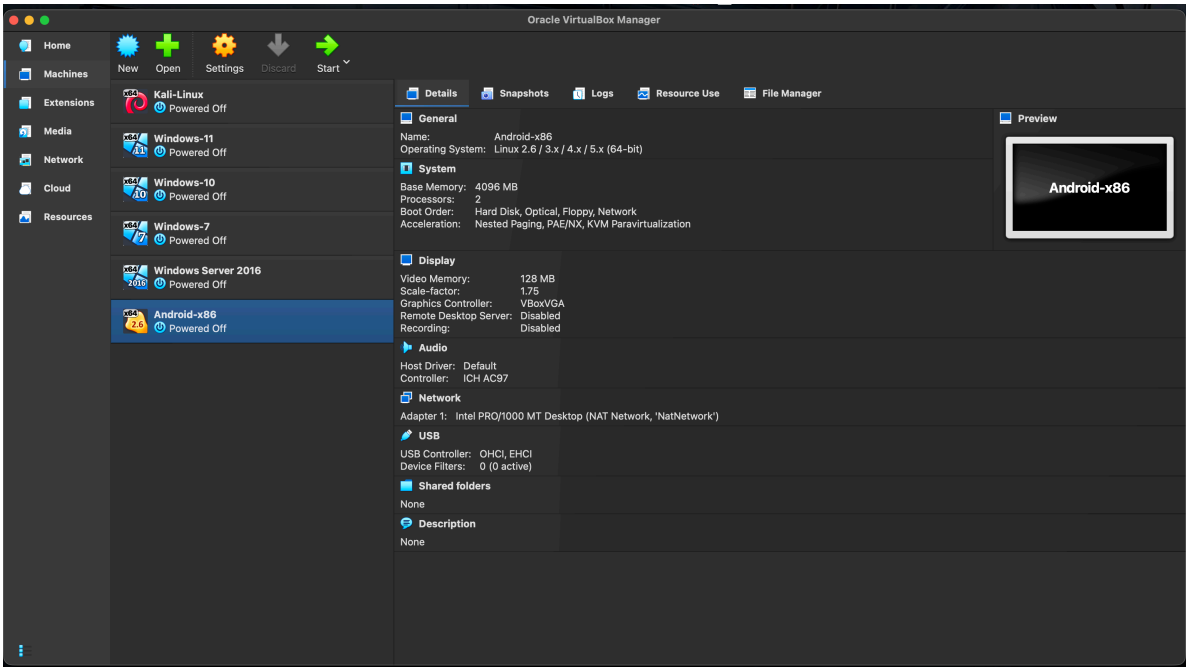
ISOLATION AND RESOURCE MANAGEMENT

Each guest operating system runs in its own virtual machine. This provides separation between guest environments while allowing controlled communication through the configured virtual network.

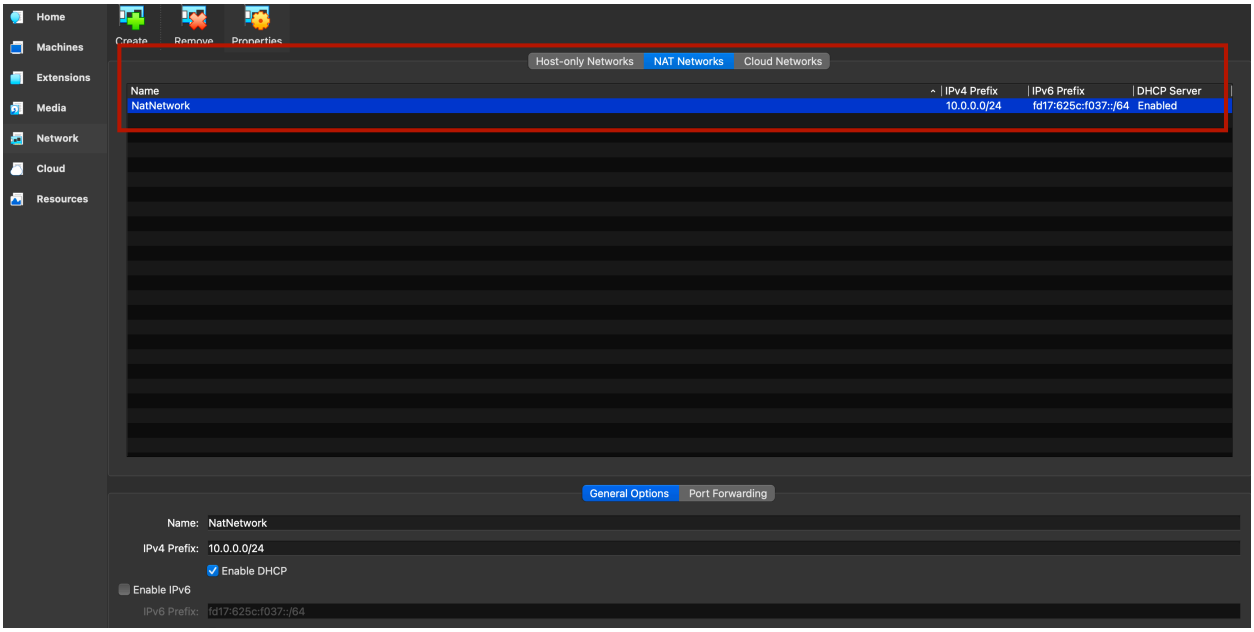
The NAT Network allows participating machines to communicate using private IPv4 addresses. Unused machines are powered off to conserve the host's physical memory and processing resources.

EVIDENCE AND POCs

VirtualBox Manager:



Network Adapter Configuration:



Network Architecture & Topology

NETWORK OVERVIEW

The task uses a shared VirtualBox NAT Network to connect six virtual machines within a common private IPv4 subnet.

The network is designed to support communication between the virtual machines while providing a NAT gateway for potential outbound network access.

Network parameter	Configuration
Network name	NATNetwork
Network type	VirtualBox NAT Network
Network address	10.0.0.0/24
Subnet mask	255.255.255.0
Default gateway	10.0.0.1
DHCP	Disabled
Address assignment	Static
Planned IP range	10.0.0.2–10.0.0.99

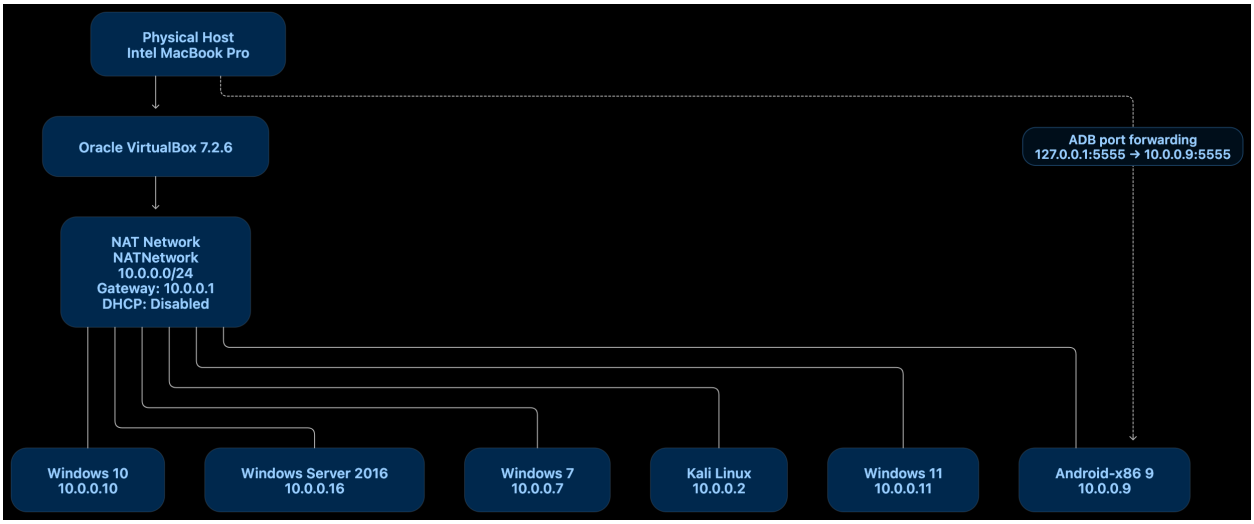
IP ADDRESS ALLOCATION

The following addressing scheme is used for the six virtual machines.

Virtual machine	Operating system	Assigned IP
Kali Linux	Kali Linux	10.0.0.2
Windows 7	Windows 7	10.0.0.7
Android	Android-x86 9.0-r2	10.0.0.9
Windows 10	Windows 10	10.0.0.10
Windows 11	Windows 11	10.0.0.11
Windows Server	Windows Server 2016	10.0.0.16

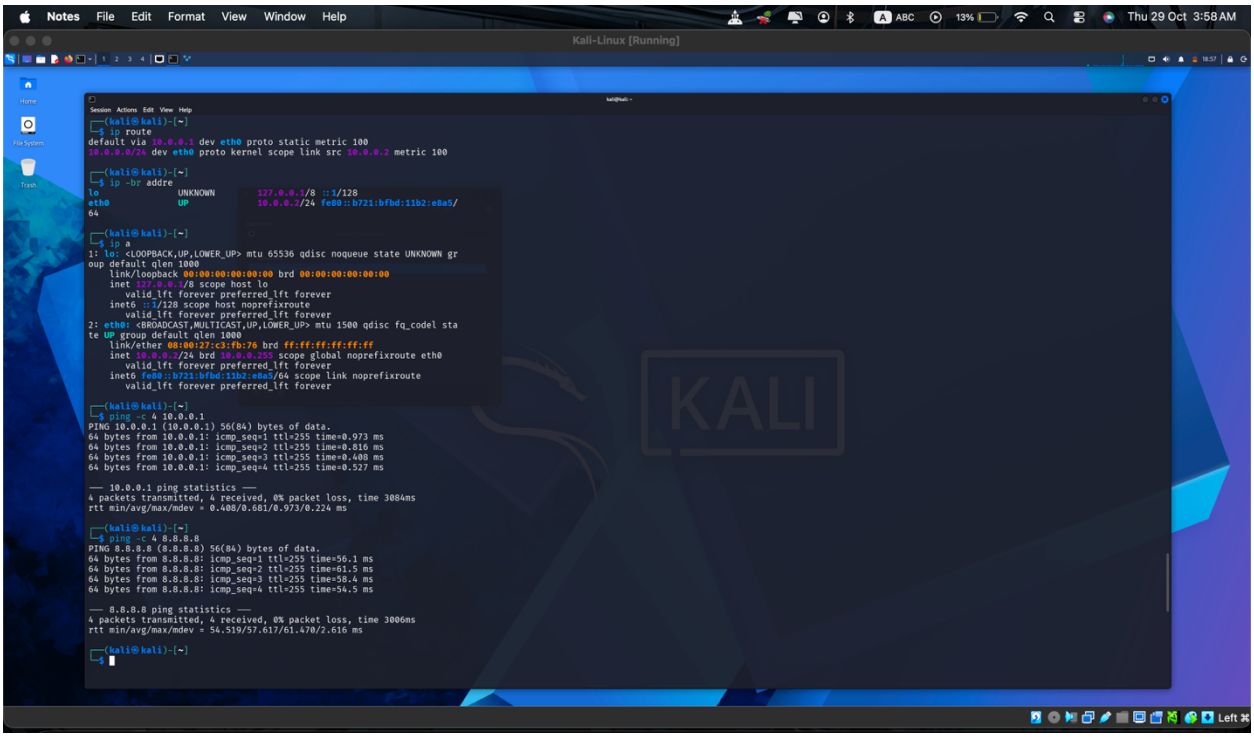
NETWORK TOPOLOGY DIAGRAM

The following diagram illustrates the logical network architecture.

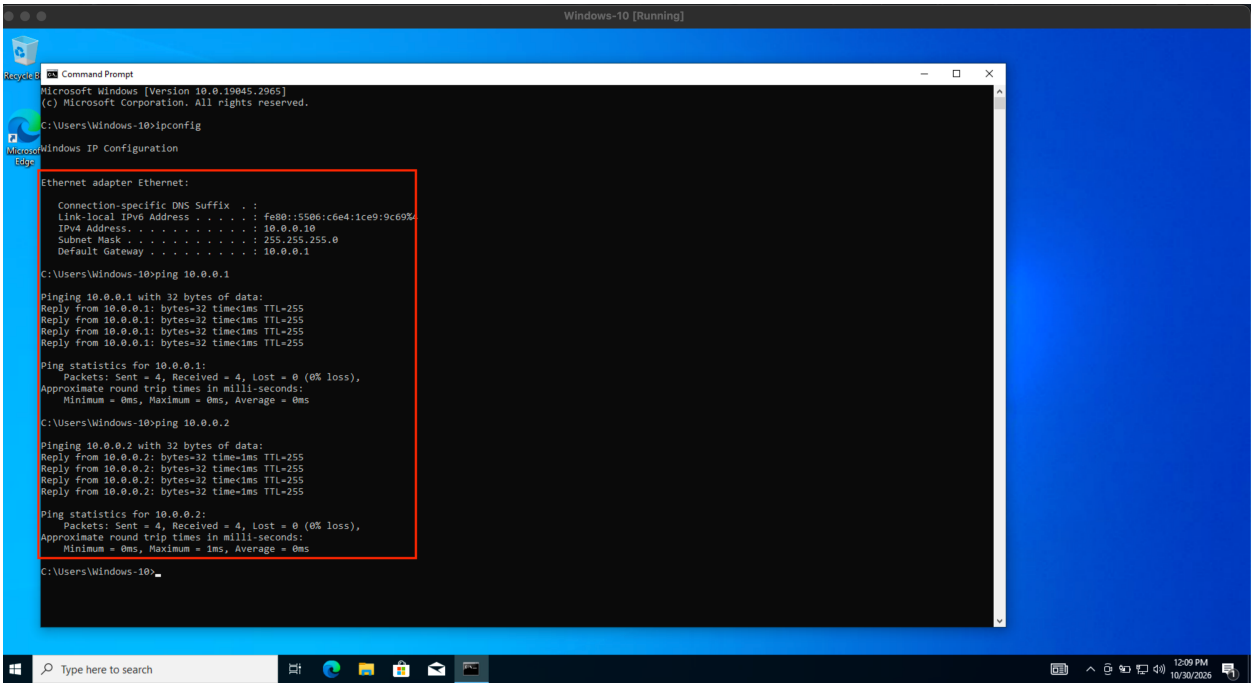


EVIDENCE AND POCs

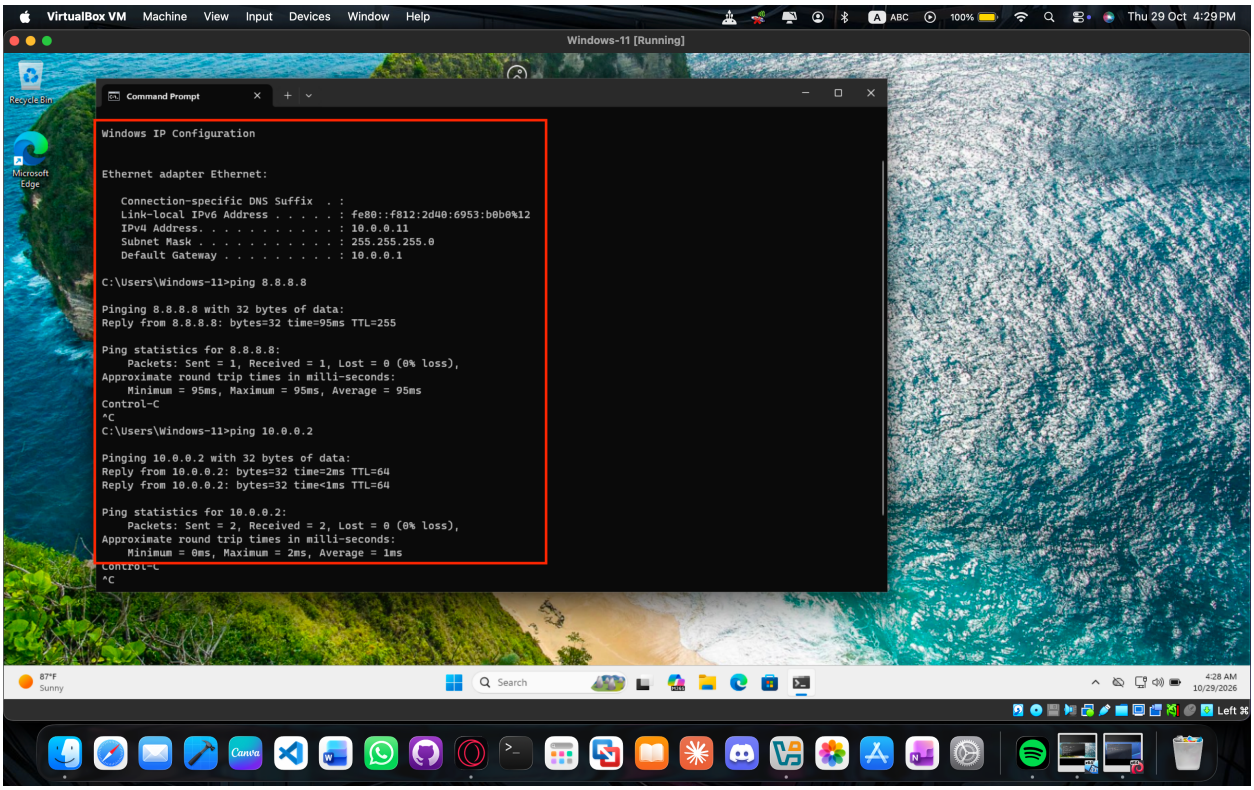
IP configuration from Kali Linux:



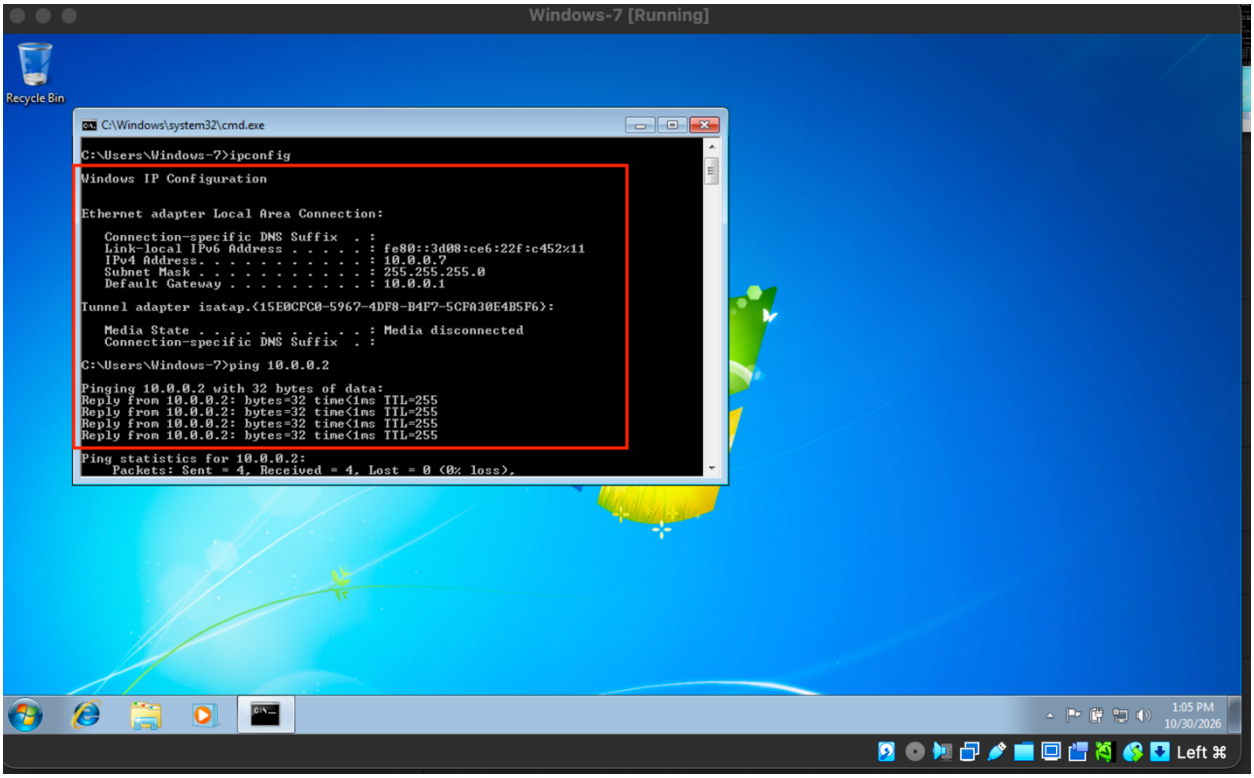
IP configuration from Windows 10:



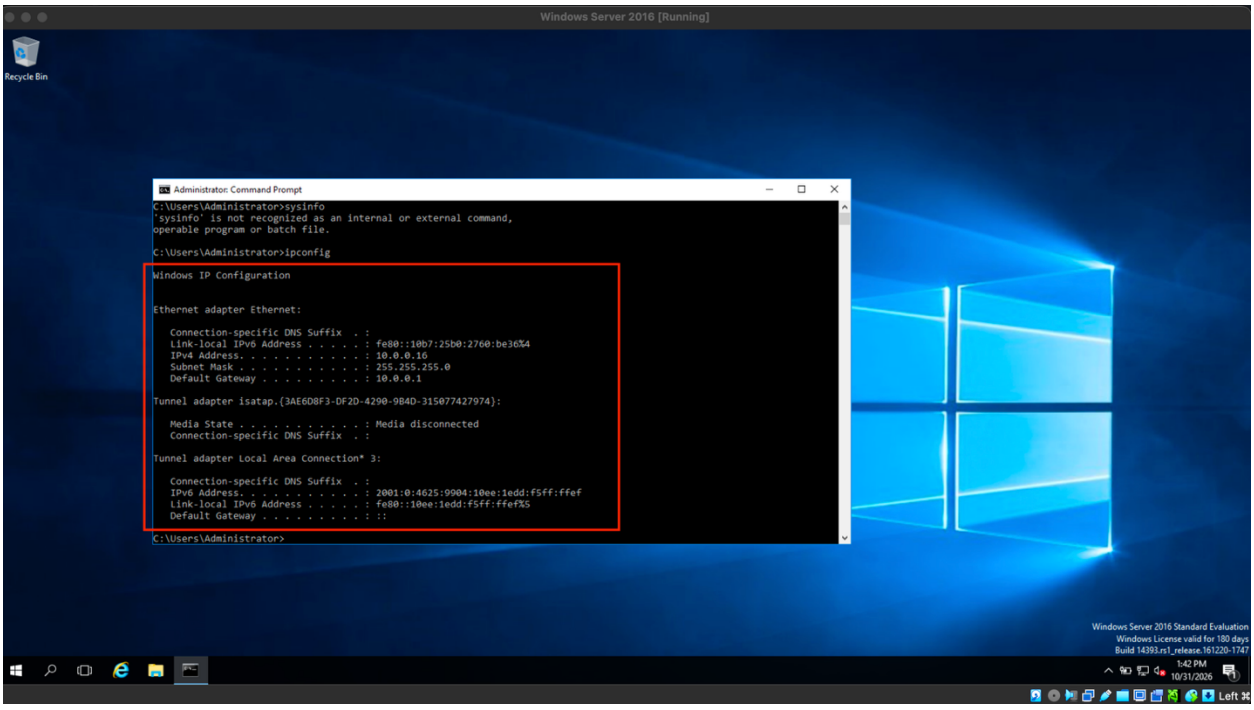
IP configuration from Windows 11:



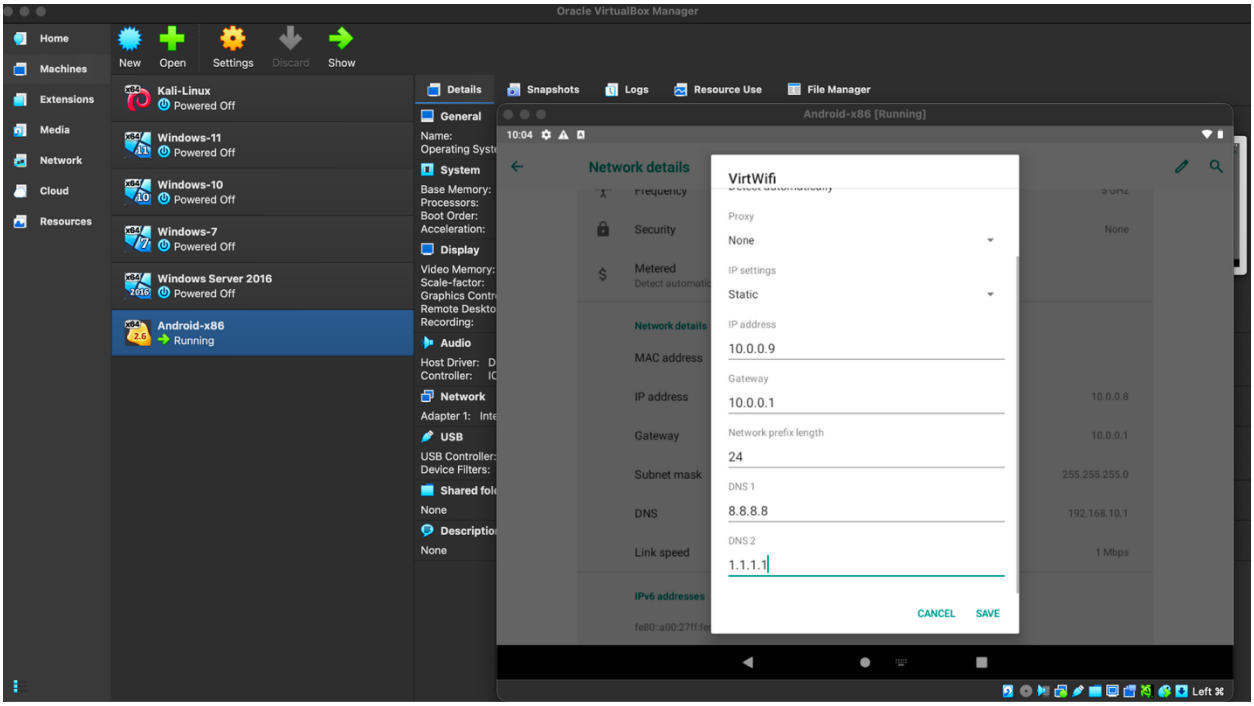
IP configuration from Windows 7:



IP configuration from Windows Server 2016:



IP configuration from Android-x86:



Virtual Machine Configuration & Operating System Deployment

OVERVIEW

This section documents the configuration and deployment of the six virtual machines created for the NETWORKWALKS-B082 cybersecurity lab.

Each virtual machine represents a different operating system environment. Together, they provide a platform for operating system administration, network communication, security testing, and practical cybersecurity exercises.

VIRTUAL MACHINE INVENTORY

No.	Virtual machine	Operating system	IP address
01	Kali Linux	Kali Linux Rolling	10.0.0.2
02	Windows 7	Windows 7	10.0.0.7
03	Android	Android-x86 9.0-r2	10.0.0.9
04	Windows 10	Windows 10	10.0.0.10
05	Windows 11	Windows 11	10.0.0.11
06	Windows Server	Windows Server 2016	10.0.0.16

OPERATING SYSTEM DEPLOYMENT DETAILS

1. Kali Linux

Purpose: Attacker and penetration-testing workstation.

IP address: 10.0.0.2

Primary tools: Nmap, Burp Suite, Wireshark, Metasploit, and other security utilities.

Kali Linux serves as the primary security testing environment for network discovery, vulnerability assessment, traffic analysis, and authorized penetration-testing exercises.

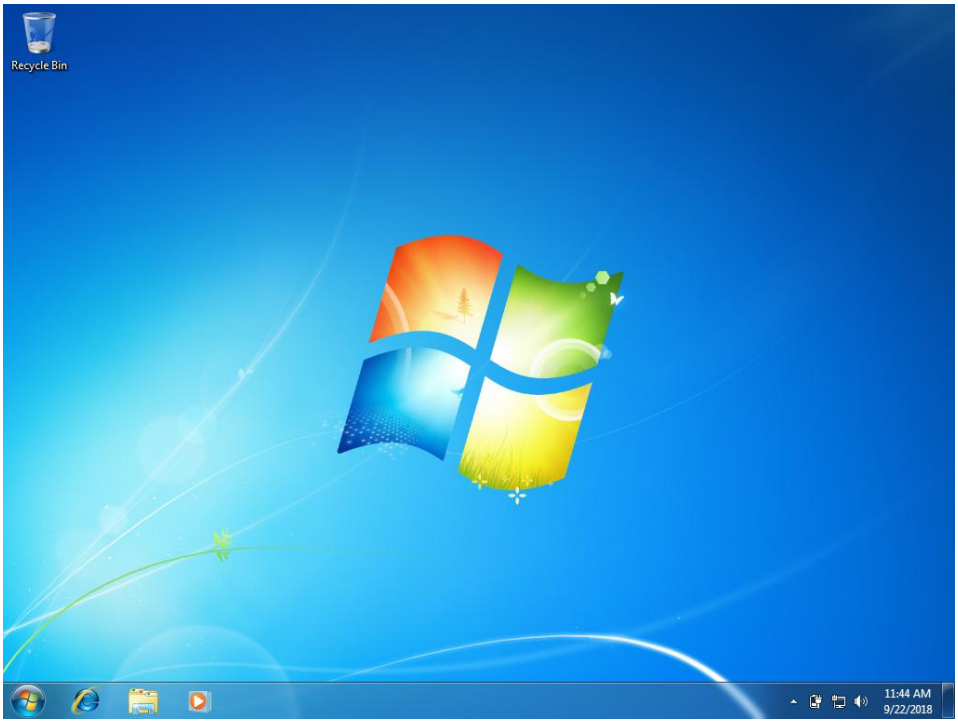


2. Windows 7

Purpose: Legacy operating system testing.

IP address: 10.0.0.7

Windows 7 provides a legacy environment for studying outdated operating systems, security configurations, network services, and compatibility issues.

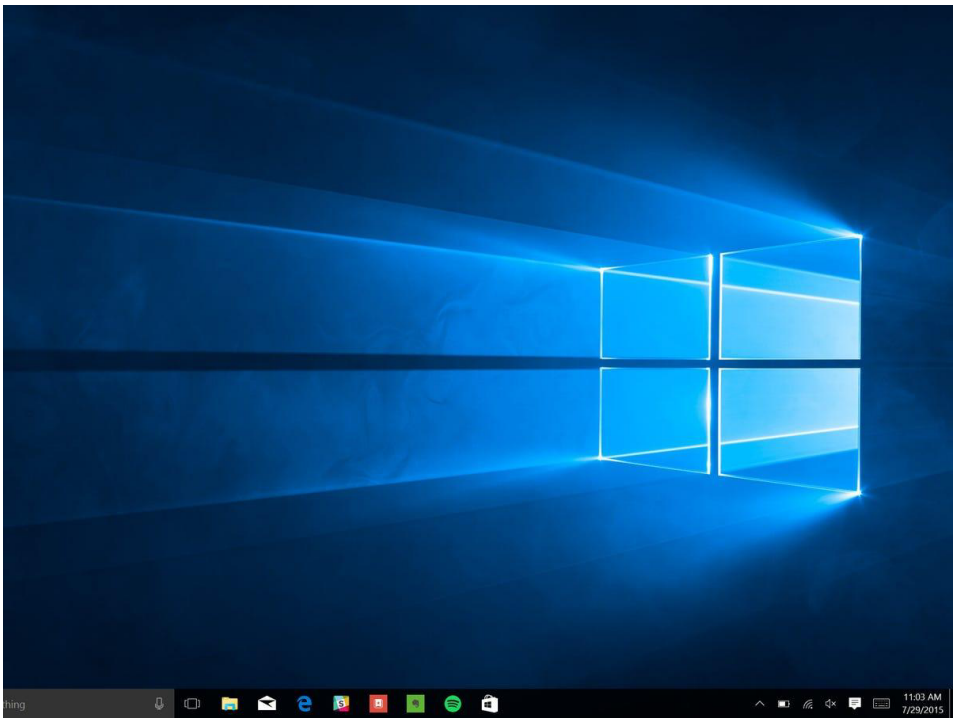


3. Windows 10

Purpose: Windows endpoint testing and administration.

IP address: 10.0.0.10

Windows 10 provides an endpoint environment for studying Windows networking, system administration, security controls, and vulnerability assessment.

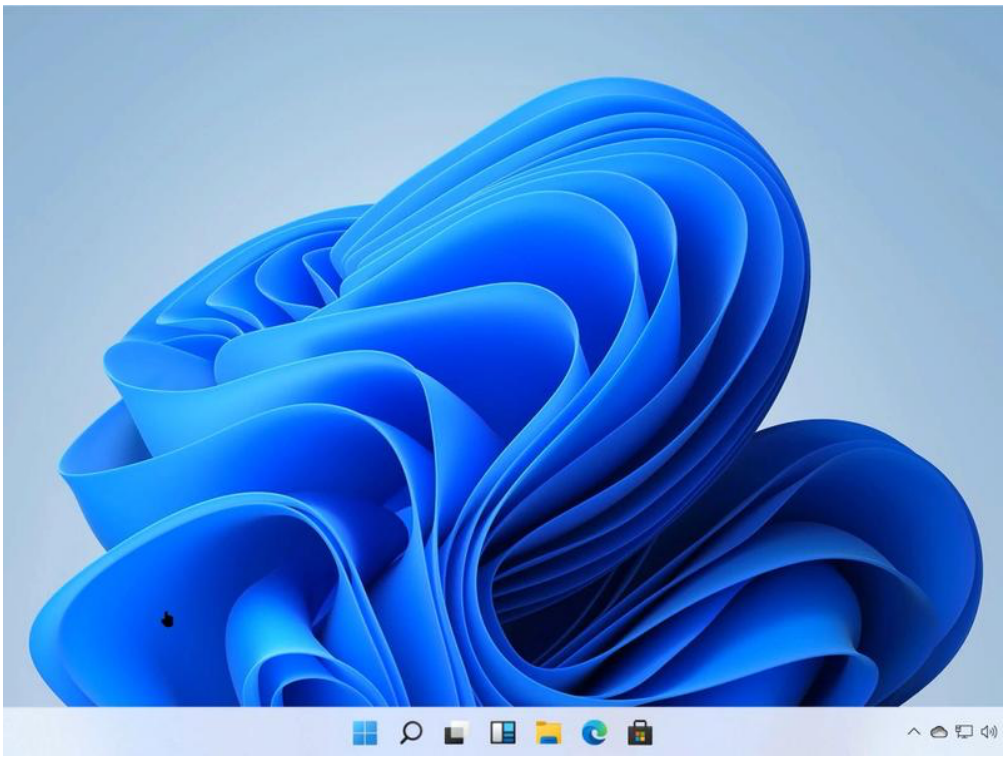


4. Windows 11

Purpose: Modern Windows endpoint testing.

IP address: 10.0.0.11

Windows 11 is included for examining modern Windows security features, network configuration, user permissions, and endpoint protection mechanisms.

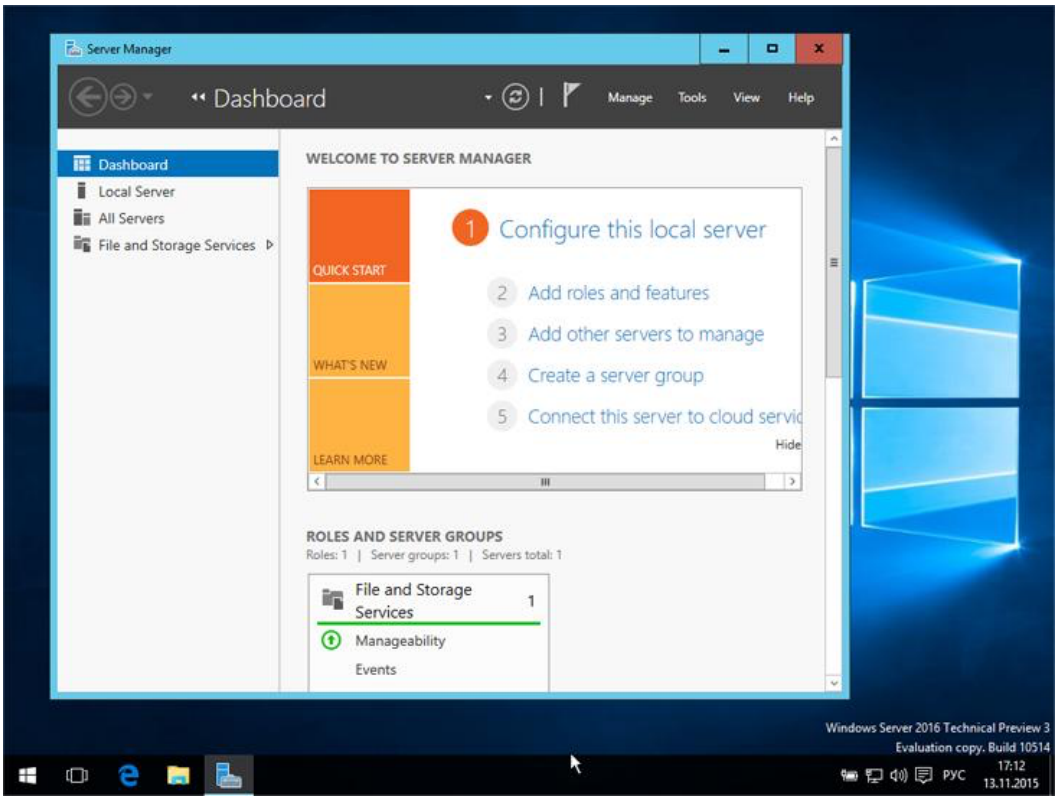


5. Windows Server 2016

Purpose: Server administration and infrastructure security.

IP address: 10.0.0.16

Windows Server 2016 provides a server environment for exploring Windows server administration, network services, access control, and infrastructure security.



6. Android-x86 9.0-r2

Purpose: Android application and mobile security testing.

IP address: 10.0.0.9

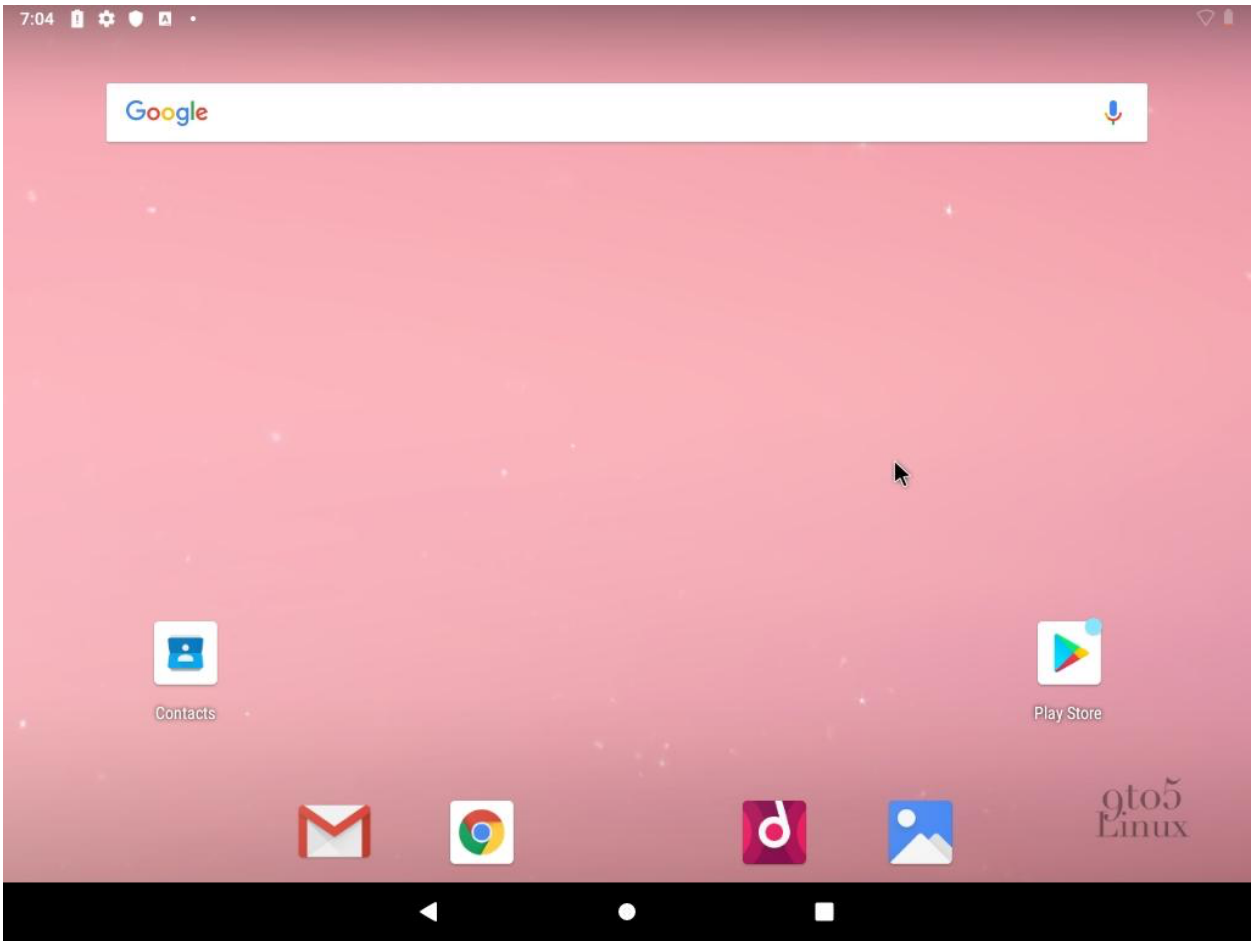
Architecture: x86_64

Android-x86 was installed on a 32 GB virtual disk with 2 GB RAM and 2 virtual CPUs.

Additional configuration:

- Android version: 9
- Android SDK: 28

- Network interface: wlan0
- ADB over TCP: Port 5555
- Root access: Available through su



Conclusion

This task successfully establishes a multi-operating-system virtualized cybersecurity lab using Oracle VirtualBox.

The environment brings together six operating systems: Kali Linux, Windows 7, Android-x86 9.0-r2, Windows 10, Windows 11, and Windows Server 2016. These systems are organized within a shared VirtualBox NAT Network using the `10.0.0.0/24` private IPv4 subnet.

The project provides a practical environment for studying network architecture, operating system administration, endpoint communication, security assessment, and mobile security testing.

Android-x86 was additionally configured for remote management through Android Debug Bridge (ADB), demonstrating host-to-guest access through VirtualBox port forwarding.

Overall, this project establishes a foundation for future cybersecurity exercises, including network reconnaissance, vulnerability assessment, traffic analysis, endpoint security testing, and penetration-testing simulations.

Learning Outcomes

Through this project, the following practical skills and concepts were explored:

1. Virtual machine creation and operating system deployment.
2. VirtualBox NAT Network configuration.
3. Static IPv4 addressing and subnetting.
4. Network interface and gateway configuration.
5. Inter-VM communication and connectivity testing.
6. Android-x86 installation and ADB configuration.
7. Network discovery and security testing preparation.
8. Technical documentation and evidence collection.

References

OFFICIAL DOCUMENTATION

1. Oracle VirtualBox User Manual — Virtual machine management, virtual networking, and port forwarding.

VirtualBox Documentation

2. Kali Linux Documentation — Installation, configuration, and security tools.

Kali Linux Documentation

3. Microsoft Learn — Windows operating systems, administration, networking, and security.

Microsoft Learn

4. Android-x86 Project — Android operating system for x86-based platforms.

Android-x86 Official Website

5. Android Debug Bridge (ADB) — Android device communication and debugging.

Android Developers — ADB

7.2 Networking and Security Resources

6. Nmap Reference Guide — Network discovery and port scanning.

Nmap Reference Guide

7. Wireshark Documentation — Network protocol analysis and packet inspection.

Wireshark Documentation

8. OWASP Web Security Testing Guide — Web application security testing methodology.

OWASP WSTG

9. OWASP Top 10 — Common web application security risks.

OWASP Top 10

10. Metasploit Documentation — Penetration-testing framework documentation.

Metasploit Documentation